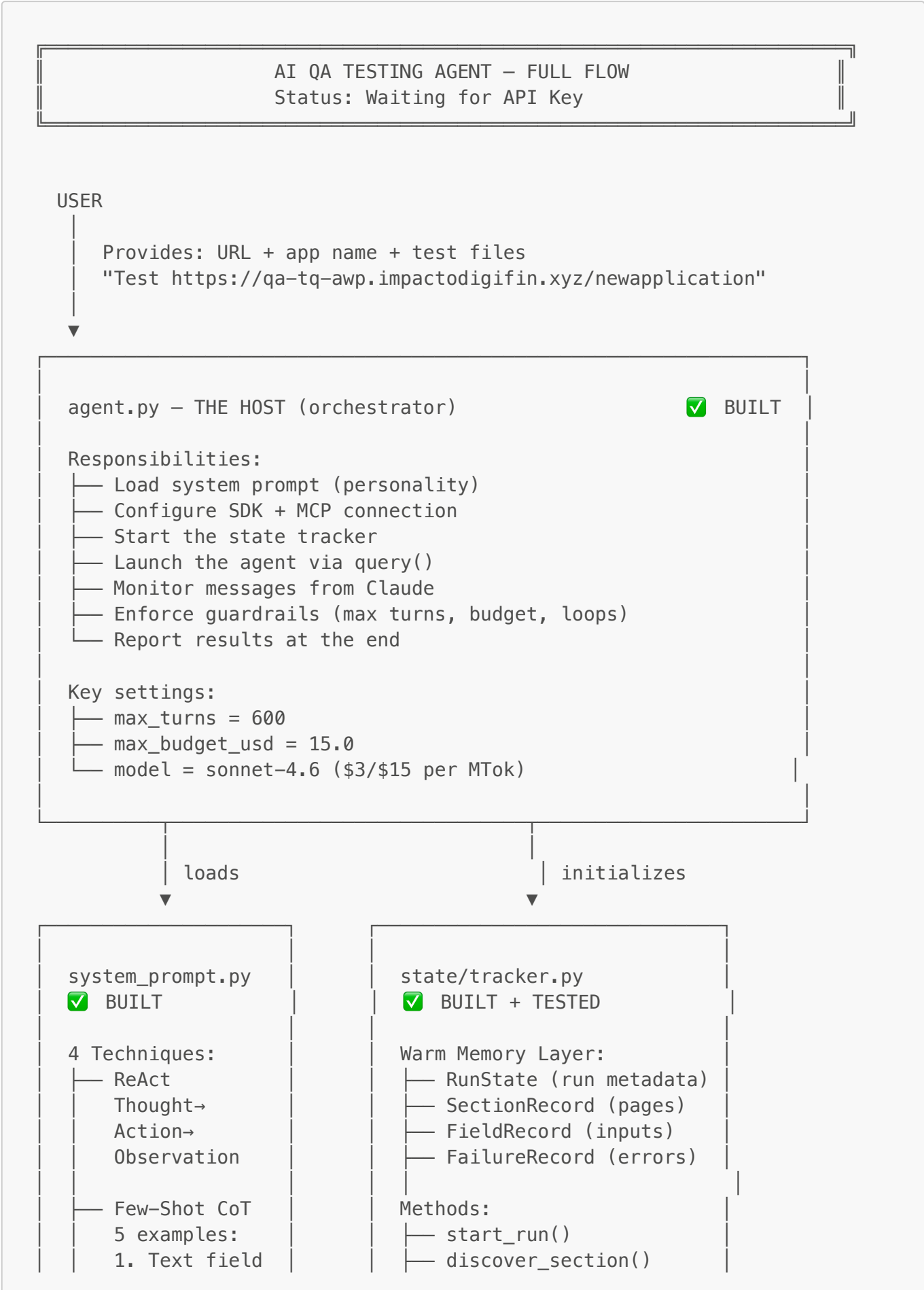
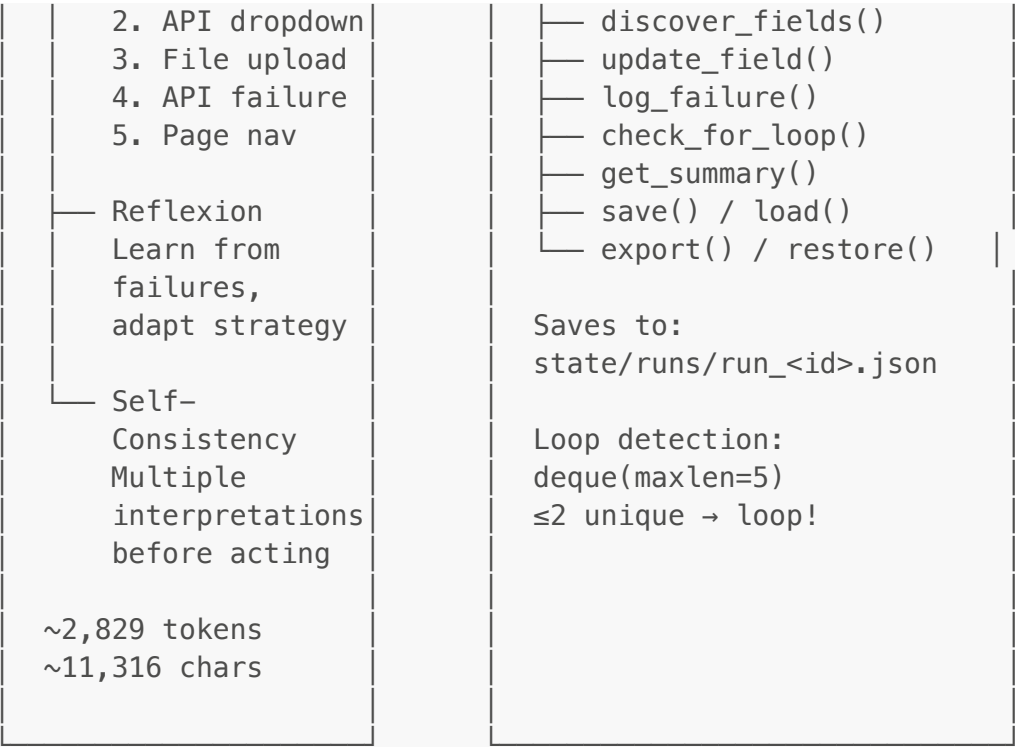


AI QA Testing Agent — Architecture Flow





agent.py calls query() — 🕒 NEEDS API KEY

▼

Claude Agent SDK (CLIENT)

✅ INSTALLED

pip: claude-agent-sdk v0.1.44

What it does:

Sends system prompt + user message to Claude API

Receives Claude's response (text + tool calls)

Executes tool calls on the MCP server

Sends tool results back to Claude

Repeats until max_turns or Claude says "done"

Streams messages back to agent.py

You don't build this – Anthropic built it

stdio transport (subprocess pipes)

"type": "stdio" in config

▼

Chrome DevTools MCP (SERVER)

✅ INSTALLED

npm: chrome-devtools-mcp v0.18.1

26 Tools exposed:

Navigation

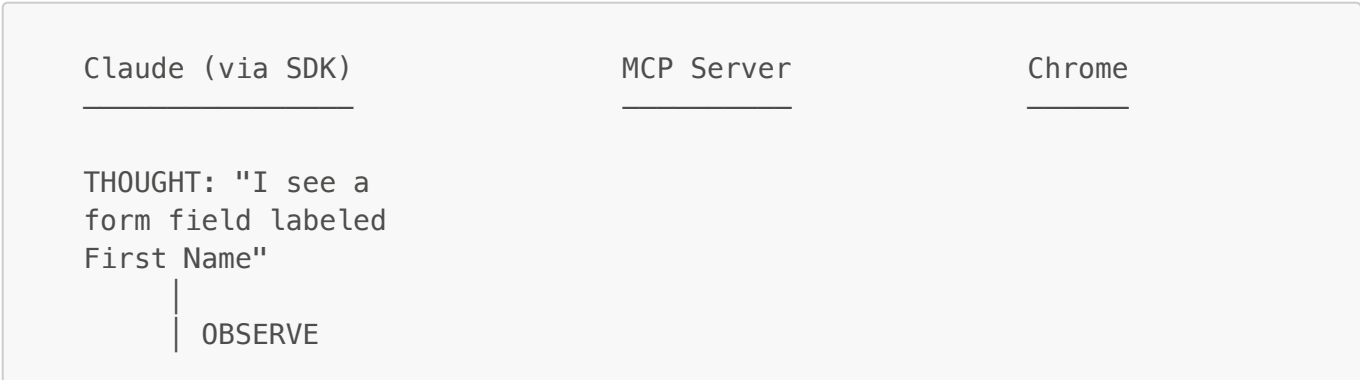
Observation

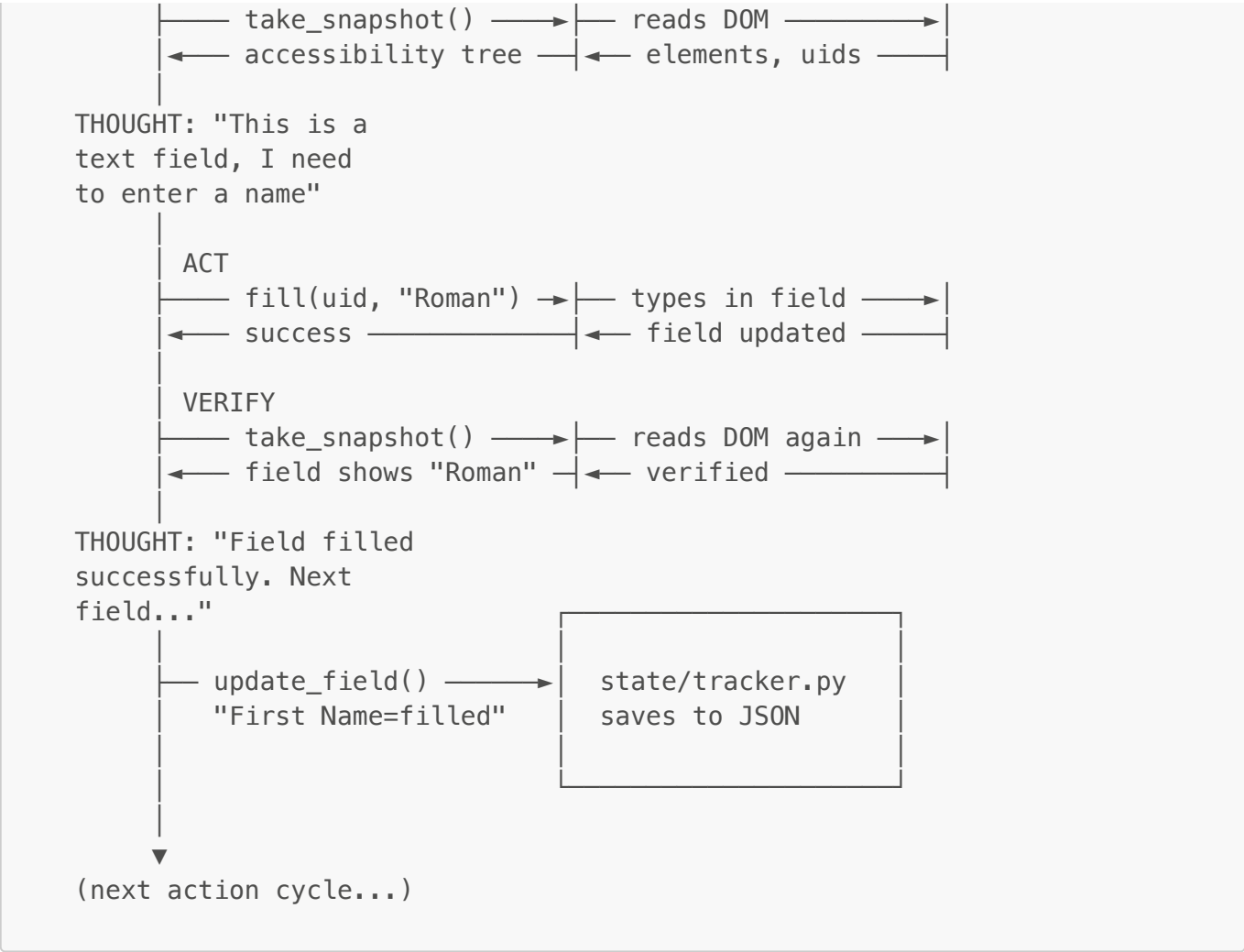
Interaction



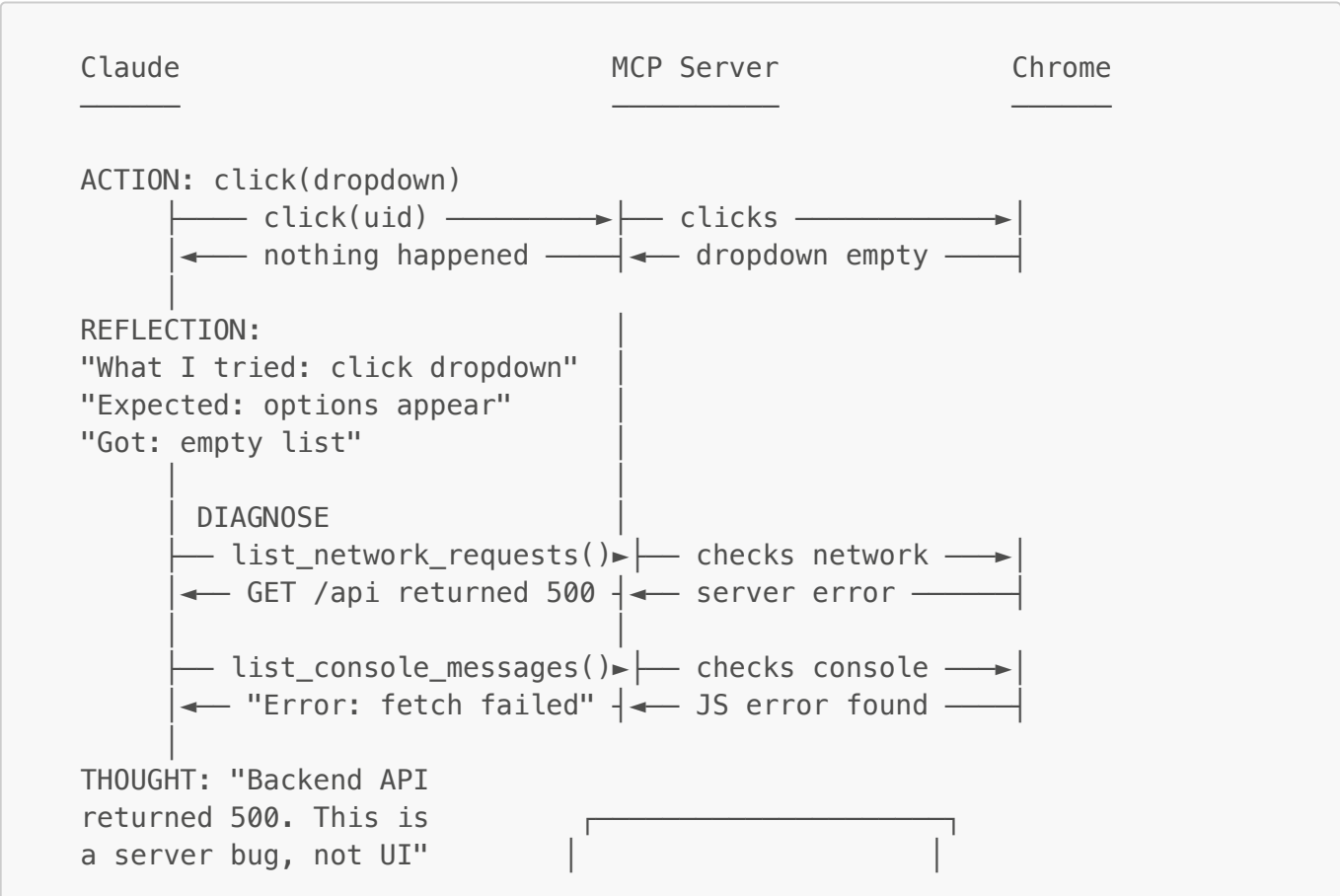
One Action Cycle

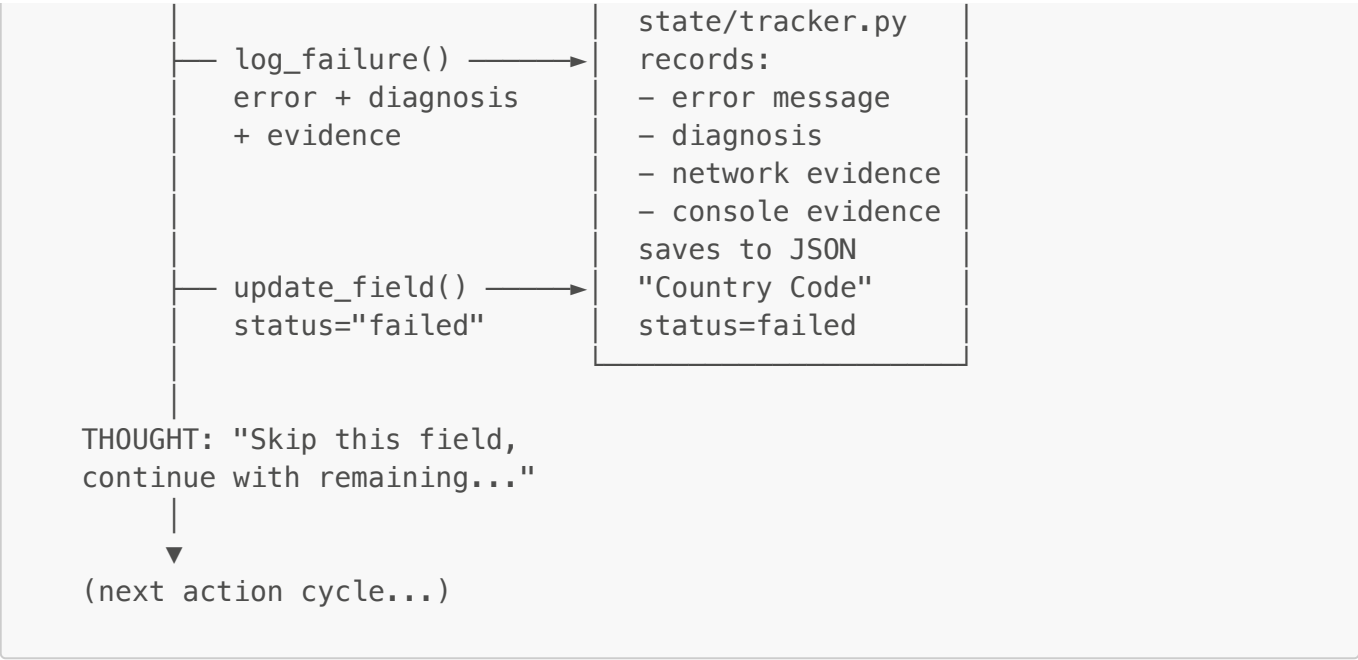
This repeats 350-500 times per full run.





Failure Cycle (when something breaks)

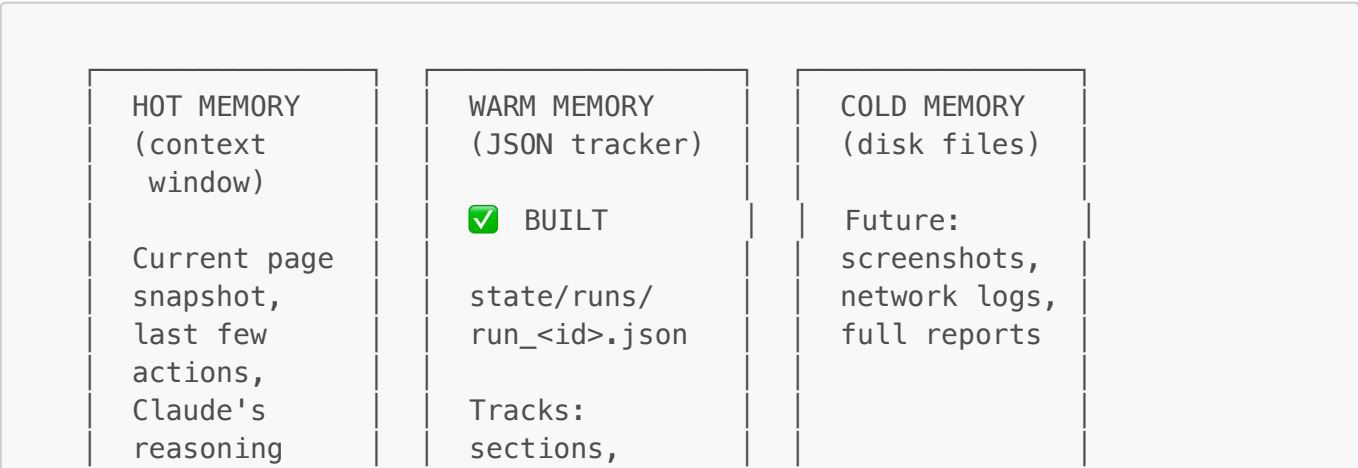




Guardrails (safety nets)

GUARDRAIL	TRIGGER	ACTION
Max turns	turn > 600	Stop run
Max budget	cost > \$15	Stop run
Loop detection	≤2 unique in last 5 actions	Skip element, move on
Max retries	3 fails on same element	Log + skip
Consecutive failures	5 in a row	Stop section
Unexpected page	CAPTCHA, 404, login screen	Stop + report
Browser crash	Chrome dies	Save state, try restart

Three-Tier Memory



~200K tokens limit Managed by SDK (auto)	fields, failures, progress Managed by tracker.py	Managed by agent.py
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